(i)

Let $u = 1 - x, \quad dx = -du,$

x	0	1
u	1	0

$$\begin{aligned} \mathrm{b}(p,q) &= \int_0^1 x^{p-1}(1-x)^{q-1}\,dx \\ &= \int_0^1 u^{q-1}(1-u)^{p-1}\,du \\ &= \mathrm{b}(q,p) \end{aligned}$$

(ii)

$$\begin{aligned} \mathrm{b}(p,q) - \mathrm{b}(p,q+1) &= \int_0^1 x^{p-1}(1-x)^{q-1}\left[1-(1-x)\right]\,dx \\ &= \int_0^1 x^{p}(1-x)^{q-1}\,dx \\ &= \mathrm{b}(p+1,q) \end{aligned}$$

$$\begin{aligned} \mathrm{b}(p+1,p) &= \mathrm{b}(p,p) - \mathrm{b}(p,p+1) \\ &= \mathrm{b}(p,p) - \mathrm{b}(p+1,p) \\ &= \frac{1}{2}\mathrm{b}(p,p) \end{aligned}$$

(iii) Let $x = \sin^2\theta, \qquad 1 - x = \cos^2\theta, \qquad dx = 2\sin\theta\cos\theta\,d\theta,$

x	0	1
θ	0	$\frac{\pi}{2}$

$$\begin{aligned} \mathrm{b}(p,q) &= \int_0^1 x^{p-1}(1-x)^{q-1}\,dx \\ &= \int_0^{\frac{\pi}{2}} (\sin^2\theta)^{p-1}(\cos^2\theta)^{q-1}(2\sin\theta\cos\theta)\,d\theta \\ &= 2\int_0^{\frac{\pi}{2}} (\sin\theta)^{2p-1}(\cos\theta)^{2q-1}\,d\theta. \end{aligned}$$

$$\begin{aligned}
\mathrm{b}(p,p) &= 2\int_0^{\frac{\pi}{2}} (\sin\theta\cos\theta)^{2p-1}\,d\theta \\
&= 2\int_0^{\frac{\pi}{2}} \left(\frac{1}{2}\sin 2\theta\right)^{2p-1} d\theta \\
&= \frac{2}{2^{2p-1}}\int_0^{\frac{\pi}{2}} (\sin 2\theta)^{2p-1}\,d\theta \\
&= \frac{1}{2^{2p-1}}\int_0^{\pi} (\sin u)^{2p-1}\,du \\
&= \frac{2}{2^{2p-1}}\int_0^{\frac{\pi}{2}} (\sin u)^{2p-1}\,du
\end{aligned}$$

When $q = \frac{1}{2}$

$$\mathrm{b}\left(p,\frac{1}{2}\right) = 2\int_0^{\frac{\pi}{2}} (\sin u)^{2p-1}\,du$$

Hence,

$$\boxed{\mathrm{b}(p,p) = \frac{1}{2^{2p-1}}\mathrm{b}\left(p,\frac{1}{2}\right)}$$

(iv)

Let $t = \dfrac{x}{1-x}$, $\qquad dx = \dfrac{1}{(1+t)^2}\,dt$, $\qquad \begin{array}{c|cc} x & 0 & 1 \\ \hline t & 0 & \infty \end{array}$

$$\begin{aligned}
\mathrm{b}(p,q) &= \int_0^1 x^{p-1}(1-x)^{q-1}\,dx \\
&= \int_0^{\infty} \left(\frac{t}{1+t}\right)^{p-1}\left(\frac{1}{1+t}\right)^{q-1}\frac{1}{(1+t)^2}\,dt \\
&= \int_0^{\infty} \frac{t^{p-1}}{(1+t)^{p+q}}\,dt
\end{aligned}$$

Hence,

$$\boxed{\mathrm{b}(p,q) = \int_0^{\infty} \frac{t^{p-1}}{(1+t)^{p+q}}\,dt}$$

(v)

$$\mathrm{b}\left(\frac{5}{2},\frac{7}{2}\right) = \int_0^{\infty} \frac{t^{\frac{3}{2}}}{(1+t)^6}\,dt \qquad \text{from part (iv)}$$

$$\mathrm{b}\left(\frac{5}{2},\frac{5}{2}\right)-\mathrm{b}\left(\frac{5}{2},\frac{7}{2}\right)=\mathrm{b}\left(\frac{7}{2},\frac{5}{2}\right)=\mathrm{b}\left(\frac{5}{2},\frac{7}{2}\right)$$

$$\begin{aligned}\mathrm{b}\left(\frac{5}{2},\frac{7}{2}\right)&=\frac{1}{2}\mathrm{b}\left(\frac{5}{2},\frac{5}{2}\right)\\&=\frac{1}{2}\cdot\frac{1}{2^4}\mathrm{b}\left(\frac{5}{2},\frac{1}{2}\right)\quad\text{from part (iii)}\\&=\frac{1}{2^5}\mathrm{b}\left(\frac{5}{2},\frac{1}{2}\right)\end{aligned}$$

$$\mathrm{b}\left(\frac{3}{2},\frac{1}{2}\right)=2\int_0^{\frac{\pi}{2}}\sin^2\theta\,d\theta=\frac{\pi}{2}\qquad\text{from part (iii)}$$

$$\mathrm{b}\left(\frac{3}{2},\frac{3}{2}\right)=\frac{1}{2^2}\mathrm{b}\left(\frac{3}{2},\frac{1}{2}\right)=\frac{\pi}{8}\qquad\text{from part (iii)}$$

$$\mathrm{b}\left(\frac{5}{2},\frac{1}{2}\right)=\mathrm{b}\left(\frac{3}{2},\frac{1}{2}\right)-\mathrm{b}\left(\frac{3}{2},\frac{3}{2}\right)=\frac{\pi}{2}-\frac{\pi}{8}=\frac{3\pi}{8}\qquad\text{from part (ii)}$$

$$\int_0^\infty\frac{t^{\frac{3}{2}}}{(1+t)^6}\,dt=\frac{1}{2^5}\cdot\frac{3\pi}{8}=\boxed{\frac{3\pi}{256}}$$

This solution was submitted by **Jon**

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